

Hardware Requirements Specification (HRS)

Project: Autonomous Solar-Powered LoRaWAN Dual Latching Valve & Multi-Channel Flow Controller Node

Prepared by: AmudarIO Research LLC

1. Project Overview & Operational Scope

The objective is to design an industrial-grade, ultra-reliable, solar-powered LoRaWAN hydraulic irrigation valve and flow monitoring controller node. The device interfaces with up to two 12V DC pulse latching solenoids, an RS485 digital water pressure sensor, two RS485 water flow meters, and additional RS485/I2C expansion sensors. It communicates bidirectionally with a central LoRaWAN gateway/ChirpStack server to execute valve open/close schedules, monitor line pressure and volumetric flow rates, and transmit battery state-of-health.

- **Component & Architectural Flexibility:** The hardware designer is free to propose alternative components, ICs, or topological enhancements based on design experience, supply availability, or thermal/power optimizations. Any significant structural or component substitutions must be submitted to and approved by AmudarIO Research LLC prior to final implementation.

2. Core Processing & RF Connectivity

- **Microcontroller:** ESP32-S3 (dual-core Xtensa LX7 processor) selected for multi-tasking valve timing, RS485 Modbus polling, fail-safe state machines, and deep sleep power management.
- **LoRa Radio:** SX1262-based LoRa transceiver module with integrated LNA (Low Noise Amplifier) for maximum link margin and sensitivity in orchard/field environments.
- **LoRa Regional Setting:** RF frontend matching optimized for the 868 MHz ISM band (EU868 standard).
- **Antenna Interface:** Standard edge-mounted SMA female connector for external chassis mounting.
- **Memory & Session Retention:** Flash Non-Volatile Storage (NVS) and RTC memory support for preserving LoRaWAN OTAA session keys, frame counters, and valve position states across reboots and deep sleep intervals.

3. Power Architecture & Battery Management

- Primary Power Input: External 12V nominal Lead-Acid battery system (10.5V to 14.8V operational range) connected via the DC LOAD terminals of an external solar charge controller.
- Input Protection: Heavy-duty reverse polarity protection (ideal diode / P-MOSFET), over-voltage clamping (TVS / Varistor), and auto-resettable PTC fuse rated for system loads.
- Internal Voltage Regulation: High-efficiency DC-DC step-down (buck) converter supplying clean, stable 3.3V power for the ESP32-S3, LoRa module, and logic circuitry.
- Battery Voltage Monitoring: High-voltage resistor divider circuit scaled for 0–18V input into an ESP32-S3 ADC pin. Must be gated via a P-channel MOSFET (high-side load switch) to prevent continuous parasitic drain from the 12V battery.
- Per-Channel Sensor Power Gating: Dedicated, switchable high-side power switches (12V / 5V / 3.3V as required) allowing the MCU to independently turn off power to each connected sensor/flow meter during sleep to minimize quiescent current.

4. Actuation Subsystem (Dual Latching Solenoids)

- Dual H-Bridge Driver: Integrated dual H-bridge motor/solenoid driver IC (or discrete MOSFET H-bridge circuits) capable of independently driving two 2-wire latching solenoid valves.
- Current & Voltage Rating: Continuous/peak capacity of at least 3.0A per channel at 12V DC to supply the short, high-energy pulses required to shift latching hydraulic plungers.
- Flyback & Inductive Surge Protection: Clamping Schottky diodes and snubber networks on all motor/valve outputs to suppress high-voltage back-EMF spikes generated during inductive turn-off.
- Control Logic: High-side and low-side drive control lines equipped with hardware pull-down/pull-up resistors and cross-conduction (shoot-through) protection to guarantee that solenoids are never accidentally energized during MCU power-up or reset.

5. RS485 Opto-Isolated Hub Architecture (Star Topology) & Sensor Ports

- 4-Channel RS485 Hub Architecture: The board must feature 4 distinct RS485 channels connected to a single hardware UART peripheral of the ESP32-S3, implemented as an active RS485 hub (star topology). Each branch port must operate in parallel, allowing the ESP32-S3 to act as the master communicating with multiple slave nodes without impedance discontinuities.
- Optoelectronic Isolation: Every RS485 channel must be optically and electrically isolated (using digital isolators/optocouplers and isolated DC-DC converters). This creates an isolation boundary that protects the MCU from ground loops, transients, and faults on any individual sensor branch.
- Channel Allocation & Power Switching:

- - Channel 1: Water Pressure Sensor port (with switchable sensor power).
- - Channel 2: Water Flow Meter 1 port. The power supply to this port must be sourced directly from the 12V battery and must be fully switchable via a positive side (high-side) transistor (P-MOSFET) or relay controlled by an ESP32-S3 GPIO.
- - Channel 3: Water Flow Meter 2 port. The power supply to this port must be sourced directly from the 12V battery and must be fully switchable via a positive side (high-side) transistor (P-MOSFET) or relay controlled by an ESP32-S3 GPIO.
- - Channel 4: Auxiliary / Expansion RS485 port (with switchable power).
- I2C Sensor Port: Dedicated 4-pin breakout connector (SDA, SCL, 3.3V, GND) with pull-up resistors for optional local temperature/humidity or digital pressure sensors.

6. User Interface, Expansion & Layout

- User Interface & Diagnostics:
 - - 1x User Push Button: Debounced tactile switch connected to an interrupt-capable GPIO for manual valve cycling, field pairing, or factory reset.
 - - 1x User-Programmable LED: Through-Hole (THT) 3mm/5mm LED or 2-pin connector for external panel visibility, driven by GPIO for status, RF feedback, and fault indications.
- Expansion Breakout: Breakout header (2.54mm pitch) exposing remaining unused GPIO pins, 3.3V, and GND.
- Layer Count & Routing: Designer's discretion (2-layer or 4-layer). Power traces for 3A solenoid current must have sufficient copper weight (1 oz or 2 oz) and trace width to prevent voltage drops. Strict physical and ground isolation must be maintained between the high-current H-bridge lines, isolated RS485 domains, and RF ground planes.
- Deep Sleep Current: Total quiescent current in standby (with ESP32-S3 in deep sleep, SX1262 in sleep mode, RS485 hub circuitry and sensor power rails gated off, and H-bridge disabled) must be $\leq 50 \mu\text{A}$.

7. Required Deliverables from Contractor

1. Native EDA design files (Altium, KiCad, or EasyEDA).
2. Complete Bill of Materials (BOM) with manufacturer part numbers and supplier links (e.g., LCSC, DigiKey).
3. Production manufacturing package: Gerber files, NC Drill files, Pick & Place (CPL) files.
4. Mechanical 3D Model: Complete .STEP assembly of the populated PCBA for enclosure fitment.
5. Bring-up test firmware validating dual H-bridge pulse generation (forward/reverse), 4-channel RS485 hub Modbus communication, per-channel 12V/5V power gating, I2C bus, battery ADC reading, button interrupts, LED toggling, and SX1262 SPI communication.